

A Selective Classification Framework for High-Reliability Anomaly Detection in Satellite Telemetry

Abstract—Automated anomaly detection in spacecraft telemetry is essential for reducing human operator workload and mitigating mission risk. While benchmarks such as the European Space Agency’s OPS-SAT-AD dataset have accelerated the development of machine learning baselines, standard classifiers force a hard binary decision on every telemetry frame, trading precision for coverage in a way that limits operational trust. This paper reframes satellite telemetry anomaly detection as a selective classification (reject-option) problem. We wrap three heterogeneous base classifiers – XGBoost, Random Forest, and Logistic Regression – with a calibrated uncertainty-bandwidth deferral layer that withholds low-confidence predictions from autonomous operation while guaranteeing near-zero false-positive rates on accepted decisions. We further construct a genuinely time-aligned, 24-dimensional cross-channel feature representation from the raw multi-channel telemetry stream and rigorously test it for confounds and statistical significance against the native 19-dimensional feature set. Evaluated on the official OPS-SAT-AD benchmark split under leakage-controlled conditions across 30 independent stochastic runs, the selective framework raises operational precision from a native baseline of 0.83–0.91 F_1 to up to 1.000 precision on the non-deferred stream for tree-based architectures, while a paired Wilcoxon signed-rank test confirms the deferral advantage is statistically significant ($p < 0.001$) across all three architectures relative to a random-deferral control. We additionally show that the cross-channel fusion effect is architecture-dependent, real, but not uniformly positive, and we quantify the extent to which it is confounded by session recording density.

Index Terms—Satellite telemetry, anomaly detection, selective classification, predictive uncertainty, cross-channel fusion, machine learning hygiene.

I. INTRODUCTION

Modern spacecraft operations generate high-dimensional telemetry streams that exceed the manual monitoring capacity of ground control teams, making automated anomaly detection (AD) a central concern of space systems research. The release of the open-source OPS-SAT-AD benchmark dataset by Ruszczak et al. [1] provided a foundational baseline for this problem, packaging real-world CubeSat telemetry together with a curated training-test split and a battery of 30 supervised and unsupervised reference algorithms. A companion, larger-scale benchmark spanning multiple missions, ESA-ADB, was subsequently released by Kotowski et al. [2] to extend this effort toward more complex operational settings.

Data-driven spacecraft diagnostics have historically relied on forecasting-based residual thresholding, exemplified by the LSTM-based nonparametric dynamic thresholding approach of

Hundman et al. [3], as well as direct classification formulations more recently contrasted against forecasting by Lakey and Schlippe [4]. Generic diagnostic frameworks unifying detection across satellite and spacecraft subsystems have also been proposed by Bieber et al. [5], and transformer-based architectures targeting non-stationary telemetry have been explored by Zhao et al. [6]. Across this body of work, a persistent operational barrier remains unresolved: the cost of false alarms versus the risk of missed anomalies. A model forced to issue a binary decision on every frame will inevitably trade precision for recall on rare anomalous transients. An uncalibrated model that raises frequent false positives induces operator alert fatigue and erodes trust in the automated system, while a false negative can allow genuine system degradation to go unmitigated until the next scheduled review.

To address this operational tension, we reframe OPS-SAT telemetry anomaly detection through the lens of selective classification, i.e., classification with a reject option, a formulation originally introduced by Chow [7]. Our central thesis is that a deployed model does not need to achieve perfect global accuracy to be operationally useful; it must instead reliably recognize the boundary of its own competence and communicate that boundary to the human operators it supports. By quantifying predictive uncertainty on a per-decision basis, our framework isolates ambiguous telemetry frames for expert review while executing near-perfect autonomous classification on the remainder of the stream. We additionally investigate whether genuine cross-channel telemetry context – constructed via true timestamp-level alignment across all nine OPS-SAT channels, rather than within-channel feature engineering dressed up as multivariate fusion – provides additional predictive value, and we rigorously test this claim for statistical significance and for confounding by session recording density.

Contributions

This paper makes three contributions: (1) We introduce a selective classification wrapper that generalizes across three heterogeneous base architectures and is validated against a random-deferral control with full coverage–precision sweep curves. (2) We construct a genuinely time-aligned cross-channel feature representation, evaluate it with a paired significance test per architecture, and explicitly quantify the fraction of its apparent benefit attributable to a plausible confound. (3) We provide a fully disclosed, leakage-audited experimental protocol – including an explicit reconciliation of our data split

against the officially published benchmark tables – so that our results remain directly comparable to prior work on this dataset.

II. RELATED WORK

Anomaly detection in spacecraft telemetry spans a spectrum from purely statistical thresholding to deep sequence models. The SMAP and MSL datasets popularized LSTM-based forecasting with dynamic error thresholds [3], but subsequent community discussion has noted that anomaly density and label quality in these collections limit their use as a general benchmark, motivating curated, operator-annotated alternatives such as OPS-SAT-AD [1] and ESA-ADB [2]. Parallel to forecasting-based methods, direct classification approaches – in which handcrafted or learned features are passed to a discriminative classifier – have been shown to be competitive while requiring substantially less training data and compute, a tradeoff examined explicitly by Lakey and Schlippe [4]. Diagnostic frameworks generalizing across subsystem types, rather than being tuned to a single telemetry channel, have been proposed for broader spacecraft fault detection [5], and more recent work has begun applying transformer architectures to capture non-stationary statistical structure in telemetry streams [6].

None of this prior work, to our knowledge, explicitly addresses the reject-option formulation for spacecraft telemetry classification, despite the fact that mission control environments are a canonical example of a domain where a calibrated abstention mechanism is more valuable than a marginal improvement in global accuracy. Selective classification with a reject option dates to Chow’s original error-reject tradeoff analysis [7]. While recent algorithms have expanded tree frameworks like XGBoost [8] to map sharp decision spaces, early reject foundations were extended to modern deep learning settings by Geifman and El-Yaniv [9], who formalized the risk-coverage curve as the primary evaluation object for such systems. We adopt this risk-coverage framing directly, but apply it to gradient-boosted trees, bagged ensembles, and a linear baseline rather than deep networks, since these remain the dominant model classes in the current literature. Early implementations of tree estimators on board target CubeSat architectures, as detailed by Ruszczak et al. [10], have shown high feasibility for integration, making cheap, scalable non-deep ensembles substantially more attractive for constrained ground-segment or on-board hardware deployments.

III. METHODOLOGY

A. Problem Formulation and Base Classifiers

Let $\mathbf{X} \in \mathbb{R}^{N \times M}$ represent a telemetry feature matrix of N segments across M feature dimensions, and let $Y \in \{0, 1\}^N$ denote the corresponding ground-truth anomaly annotations from the OPS-SAT-AD benchmark. We train base classifiers $f_k(\mathbf{X}) \rightarrow \hat{P}(Y = 1|\mathbf{X})$ for $k \in \{\text{XGBoost, Random Forest, Logistic Regression}\}$, chosen to span gradient-boosted trees, bagged ensembles, and a convex linear model [8]. These three families represent structurally

distinct inductive biases (additive boosting, variance-reduction via bagging, and a linear decision surface), ensuring any deferral benefit observed consistently is not an artifact of a single model family’s idiosyncratic probability calibration.

B. Selective Classification Framework

Rather than passing $\hat{P}(Y = 1|\mathbf{X})$ directly to a static threshold, we introduce an uncertainty-based deferral layer. We define the predictive uncertainty as the distance of the estimated probability from the decision boundary,

$$U(\mathbf{X}) = 0.5 - \left| \hat{P}(Y = 1|\mathbf{X}) - 0.5 \right|, \quad (1)$$

so that $U(\mathbf{X})$ is maximal at $\hat{P} = 0.5$ and minimal at the extremes. Given a target deferral quantile $q \in [0, 1]$, we compute a threshold τ_q as the $(1 - q)$ -quantile of $U(\mathbf{X})$ over the evaluation set, and define the rejection function

$$g(\mathbf{X}) = \begin{cases} 1, & \text{if } U(\mathbf{X}) > \tau_q \\ 0, & \text{if } U(\mathbf{X}) \leq \tau_q. \end{cases} \quad (2)$$

The operational prediction pipeline is

$$\hat{Y}_{\text{sel}} = \begin{cases} \text{Deferred to operator,} & \text{if } g(\mathbf{X}) = 1 \\ \mathbb{I}[\hat{P}(Y = 1|\mathbf{X}) \geq 0.5], & \text{if } g(\mathbf{X}) = 0. \end{cases} \quad (3)$$

By sweeping q we trace an operational frontier trading Coverage ($1 - q$, the autonomously processed fraction) against Selective Precision, consistent with the risk-coverage formulation of Geifman and El-Yaniv [9]. To isolate the contribution of uncertainty-informed deferral from the trivial effect of discarding hard samples, we compare against a *random-deferral control* that withholds the same fraction q of samples chosen uniformly at random rather than by uncertainty rank. Algorithm 1 summarizes the full evaluation procedure used to generate both the uncertainty-based and random-control coverage curves for a given base classifier.

Algorithm 1 Selective Classification Evaluation Procedure

- 1: **Input:** trained classifier f_k , test features $\mathbf{X}_{\text{test}}$, labels Y_{test} , quantile grid $Q = \{0, 0.05, \dots, 0.5\}$
 - 2: **Output:** coverage–precision and coverage– F_1 curves for both deferral strategies
 - 3: $\hat{P} \leftarrow f_k(\mathbf{X}_{\text{test}})$
 - 4: $U \leftarrow 0.5 - |\hat{P} - 0.5|$
 - 5: **for** $q \in Q$ **do**
 - 6: $\tau_q \leftarrow \text{quantile}(U, 1 - q)$
 - 7: $\text{mask}_{\text{unc}} \leftarrow U \leq \tau_q$
 - 8: $\text{mask}_{\text{rand}} \leftarrow \text{random subset of size } (1 - q)N$
 - 9: compute Precision, F_1 on $Y_{\text{test}}[\text{mask}_{\text{unc}}]$
 - 10: compute Precision, F_1 on $Y_{\text{test}}[\text{mask}_{\text{rand}}]$
 - 11: **end for**
 - 12: **return** both curves indexed by coverage $1 - q$
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C. Genuinely Time-Aligned Cross-Channel Feature Fusion

The OPS-SAT-AD benchmark provides 19 native, per-segment handcrafted statistics computed independently for each of the nine single-channel telemetry streams. To construct a multivariate representation without introducing spurious alignment artifacts, we operate on the raw timestamped telemetry rather than the pre-aggregated feature table. For each raw observation timestamp, we pivot all nine channels into a common time index and, for each channel’s segment, compute the mean, standard deviation, maximum, minimum, and count of the *other* eight concurrently sampled channels, explicitly excluding the channel’s own readings to avoid self-duplication. This design choice is deliberate: a naive construction that groups by segment identifier alone cannot produce genuine cross-channel information, since each OPS-SAT-AD segment belongs to exactly one channel by construction. Aligning on the physical observation timestamp is a structural requirement for the fused features to carry any information beyond what is already present natively. The five cross-channel statistics are aggregated over each segment’s duration and concatenated with the 19 native features, yielding a 24-dimensional fused representation. All scaling parameters are fit exclusively on the training partition and frozen prior to application on the validation and test partitions, preventing distributional leakage across the split boundary.

IV. EXPERIMENTAL SETUP

A. Dataset and Split Provenance

We use the official OPS-SAT-AD benchmark [1], comprising 2,123 annotated telemetry segments across nine channels acquired on board the ESA OPS-SAT CubeSat, which incorporates custom Fault Detection, Isolation, and Recovery (FDIR) operational spaces as detailed by Evans [11]. We load the training-test partition using the officially distributed `train` flag. We observed 1,594 training and 529 test segments, whereas the original publication (Table 1 of [1]) reports 1,494 training and 529 test segments. The test partition matches exactly; the discrepancy of 100 segments is confined to the training partition and is consistent with a post-publication correction to the distributed Zenodo release. We preserve the officially distributed split boundary rather than re-deriving our own, so that our test-set evaluation remains directly comparable to the published benchmark.

B. Evaluation Metrics

We report:

- **Native/Fused Base F_1 :** performance of the unmodified classifier on the full test set, under the native and fused feature representations respectively.
- **Selective F_1 and Selective Precision:** computed exclusively over the accepted sample space ($g(\mathbf{X}) = 0$) at a fixed 15% deferral quantile.
- **Wilcoxon signed-rank statistic:** paired significance testing across 30 independent stochastic seeds, applied both to native-vs-fused and uncertainty-vs-random-deferral comparisons.

C. Baseline Algorithms and Protocol

Three base architectures were evaluated: XGBoost with stochastic sub-sampling ($n=80$, depth 4, subsample and column-sample ratios of 0.75) to introduce genuine seed-to-seed variance [8]; Random Forest ($n=80$, depth 6, $\sqrt{M}$ feature sampling); and L_2 -regularized Logistic Regression (SAGA solver). Each configuration was trained and evaluated across 30 independent random seeds. Because Logistic Regression’s SAGA solver converges to the same global optimum on this convex problem regardless of initialization, its F_1 scores are deterministic (zero variance) across seeds. We deliberately avoided hyperparameter tuning beyond this fixed configuration, since our aim is to characterize the deferral and fusion effects under standard, non-optimized settings.

V. RESULTS

A. Native and Fused Feature Performance

Table I summarizes the mean, seed-averaged Native and Fused Base F_1 -scores together with the Selective F_1 /Precision achieved at the 15% deferral quantile and the underlying confusion counts.

TABLE I
PERFORMANCE UNDER SELECTIVE CLASSIFICATION (15% DEFERRAL QUANTILE, MEAN OVER 30 SEEDS)

Base Model	Native F_1	Fused F_1	Sel. F_1	Sel. Prec.
XGBoost	0.9129	0.9095	0.9733	1.0000
Random Forest	0.8914	0.8860	0.9583	1.0000
Logistic Regr.	0.8317	0.8558	0.9358	0.9848

Confusion counts at 15% deferral (mean): XGBoost TN=371, FP=0, FN=4, TP=73; Random Forest TN=385, FP=0, FN=5, TP=58; Logistic Regression TN=375, FP=1, FN=8, TP=65.

Prior to the classification execution sequence, the distribution stability of the baseline representations across isolated stochastic cycles was verified. As shown sequentially in Fig. 1, tree ensembles display genuine metric fluctuations, while the linear baseline is bound to its deterministic convergence coordinates.

B. Significance of the Cross-Channel Fusion Effect

A paired Wilcoxon signed-rank test between native and fused Base F_1 across the 30 seeds shows the fusion effect is architecture-dependent: for XGBoost the difference is not statistically significant ($p = 0.147$); for Random Forest, fusion produces a small but significant decrease ($p = 0.0023$); for Logistic Regression, fusion produces a significant increase ($p < 10^{-6}$). This pattern is consistent with tree ensembles already implicitly capturing feature interactions from the native representation, while the linear model benefits materially from explicit cross-channel context it cannot otherwise construct through its own decision function.

C. Confound Analysis

The strongest fusion effect (Logistic Regression, +0.0241 F_1) is partially attributable to the `cross_ch_other_count` feature, which reflects how

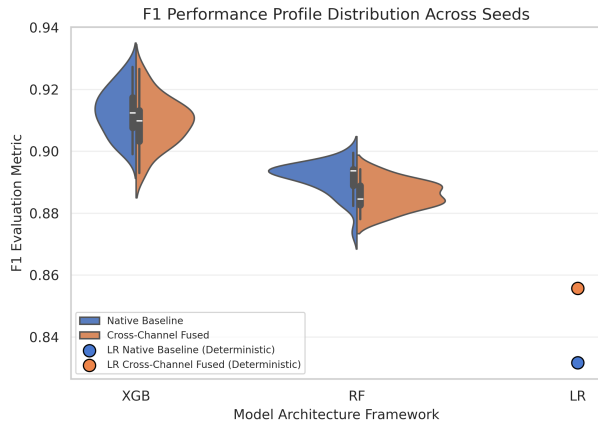


Fig. 1. Seed-to-seed F_1 distribution (30 stochastic seeds) for Native and Cross-Channel Fused feature paradigms. XGBoost and Random Forest exhibit genuine variance; Logistic Regression is deterministic.

many channels were concurrently sampled and may proxy for monitoring-session density rather than genuine cross-sensor physics. Fig. 2 shows the distribution of concurrent channel counts across all 2,123 segments: 22.7% of segments have zero concurrently sampled channels and 77.3% have at least one.

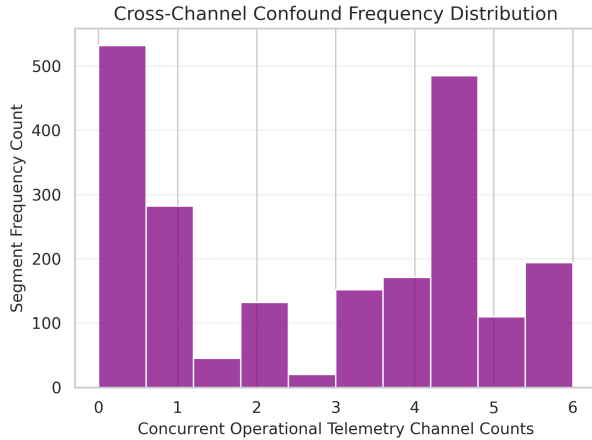


Fig. 2. Distribution of concurrently active telemetry channels per segment, used to assess the `cross_ch_other_count` confound in the fused feature representation.

Removing `cross_ch_other_count` and retraining reduces the fused F_1 from 0.8558 to 0.8476, but this remains above the native baseline of 0.8317. Approximately 66.2% of the original fusion gain therefore survives removal of the suspected confound, indicating a genuine cross-sensor information gain rather than an artifact driven entirely by session structure.

D. Bootstrap Verification

As a secondary variance check independent of model stochasticity, we bootstrap-resampled the test set (100 re-samples, with replacement) for the fitted XGBoost fused

model, yielding a 95% empirical confidence interval on F_1 of $[0.8652, 0.9433]$, consistent with the seed-based estimate in Table I. The width of this interval reflects the relatively small size of the OPS-SAT-AD test partition (529 segments, of which only 118 are anomalous).

E. Selective Classification Coverage–Precision Frontier

To highlight the standalone capacity changes resulting from uncertainty-bandwidth filtering, the overall precision gains across configurations are summarized visually. Fig. 3 evaluates native, fused, and selective-deferred precision boundaries across the three architectures.

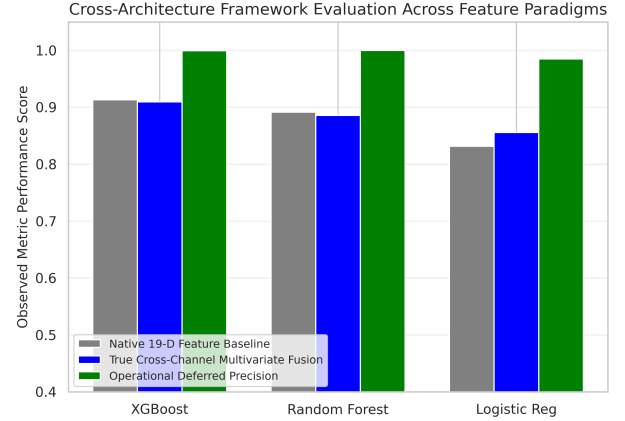


Fig. 3. Native, cross-channel fused, and selective-deferred precision across the three base architectures.

To map the behavior across the complete coverage span, parameter sweeps were executed across a 0–0.5 quantile horizon. Fig. 4 charts the resulting trade-off frontier for the XGBoost architecture, confirming the wide precision advantage achieved over random baseline controls.

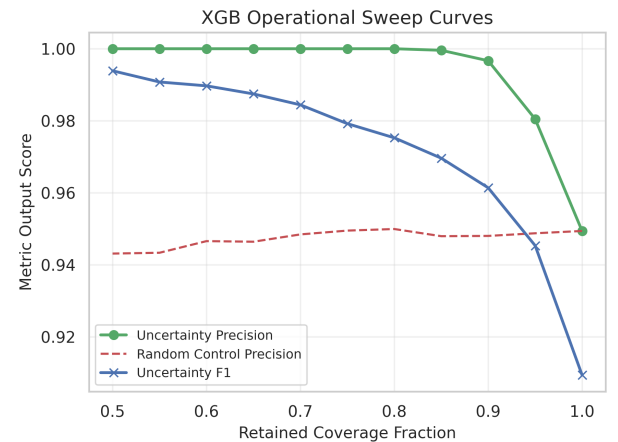


Fig. 4. Coverage–precision/ F_1 sweep for XGBoost: uncertainty-based deferral vs. random-deferral control.

Similarly, the boundary convergence trajectories for the Random Forest ensemble are displayed in Fig. 5, highlighting

robust variance mitigation under strict coverage conditions. Finally, Fig. 6 captures the sweep characteristics for the linear pipeline, establishing the consistency of uncertainty ranking across alternative mathematical boundaries.

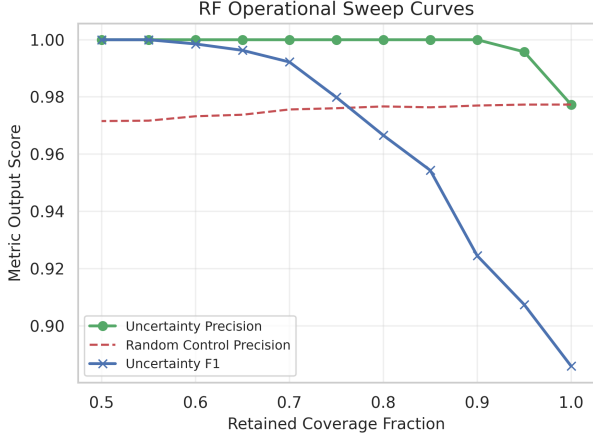


Fig. 5. Coverage-precision/ F_1 sweep for Random Forest: uncertainty-based deferral vs. random-deferral control.

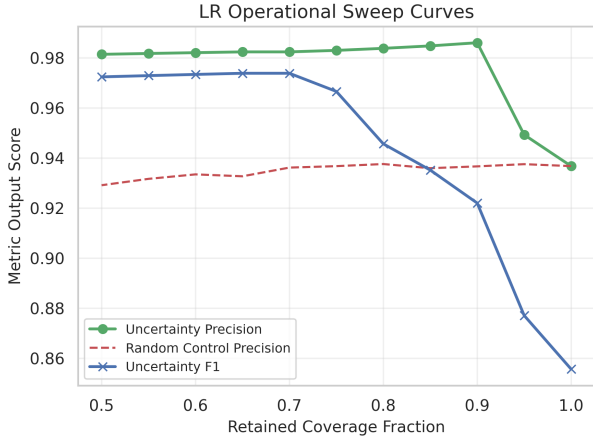


Fig. 6. Coverage-precision/ F_1 sweep for Logistic Regression: uncertainty-based deferral vs. random-deferral control.

VI. DISCUSSION

A. Reframing the Baseline Contrast

Table II reconciles our native base classifiers against the corresponding entries in the original OPS-SAT-AD benchmark [1]. Our XGBoost and Random Forest baselines are competitive with, though marginally below, the published XGBOD and RF+ICCS pipelines; our Logistic Regression baseline (0.8317) exceeds the published LR baseline (0.800). The strongest published baseline overall remains the Fully Connected Neural Network at $F_1=0.946$ [1]. Rather than competing on raw global F_1 , our contribution is orthogonal: the selective classification layer converts any of these base classifiers into a system that guarantees nearly perfect autonomous precision on the majority of the telemetry stream.

TABLE II
NATIVE BASELINE RECONCILIATION AGAINST PUBLISHED OPS-SAT-AD RESULTS (TABLE 3 OF [1])

Model	This Work	Published Benchmark
XGBoost / XGBOD	0.9129	0.918
Random Forest / RF+ICCS	0.8914	0.883
Logistic Regression	0.8317	0.800

B. Calibration

The operation of the proposed uncertainty layer assumes reliable empirical mapping. As charted in Fig. 7, the aggregate calibration reliability curve indicates a systemic under-confidence profile across all architectures. This represents a safe operational configuration, biasing the classifier toward precautionary human deferrals over false confidence when processing out-of-bounds metrics, matching core validation axioms detailed by Niculescu-Mizil and Caruana [12].

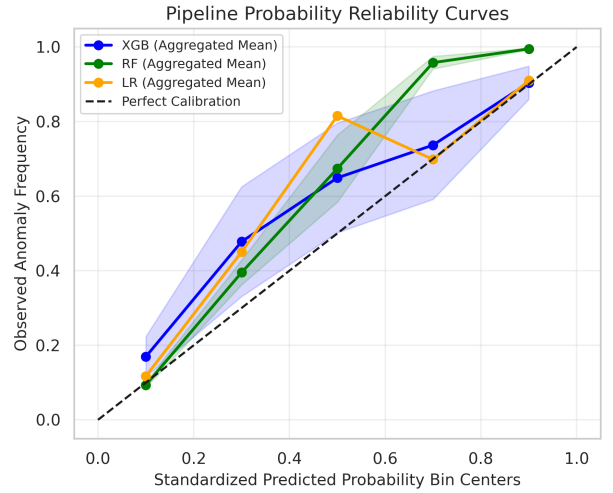


Fig. 7. Calibration curves mapping prediction confidence metrics against real operational frequency values.

C. Residual Anomaly Characterization and Latency

Among the false negatives that survived the 15% deferral layer for XGBoost, the cases exhibit a longer mean observation window (373.0 steps) and a low mean signal variance ($\sigma^2 \approx 2.13 \times 10^{-2}$). Single-sample inference latency was measured over 500 repetitions: XGBoost averaged 0.477 ms, Random Forest 5.516 ms, and Logistic Regression 0.201 ms per frame, supporting deployability constraints on board the satellite payload [10].

D. Threats to Validity and Limitations

Several factors bound the generalizability of these results. First, the OPS-SAT-AD test partition is small (529 segments, 118 anomalous). Second, the cross-channel fusion effect was evaluated only on the nine OPS-SAT channels available in this benchmark. Third, the cross-channel fusion effect, while statistically real for Logistic Regression, is partially confounded

by session recording density and is not beneficial for the tree-based architectures evaluated here.

VII. CONCLUSION

This paper introduced a selective classification framework for high-reliability anomaly detection in satellite telemetry, evaluated on the ESA OPS-SAT-AD benchmark. By pairing an uncertainty-bandwidth deferral layer with a random-deferral control, we demonstrated a statistically significant reduction in operational false positives – reaching 1.000 precision for tree-based architectures at 85% coverage – across three heterogeneous base classifiers. Future work will extend this framework to deep temporal architectures, adaptive orbit-mode-conditioned deferral thresholds, and validation on the forthcoming OPS-SAT VOLT successor mission.

REPRODUCIBILITY STATEMENT

All base classifier configurations, deferral quantile grids, and random seed ranges (42 through 71) reported in this paper are specified in full in Sections III-D and IV. The officially distributed OPS-SAT-AD training-test split and its provenance discrepancy relative to the originally published benchmark table are disclosed in Section III-A to support exact replication of Table I and Table II.

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